1.4 continuity

To Define Continuity @ a point:
(all 3 must be true, has to be exact)

1) $f(c)$ is defined   NEx: {VA, hole}

2) $\lim_{x \to c} f(x)$ must exist   NEx: {jump, VA}

3) $f(c) = \lim_{x \to c} f(x)$ NEx:

Practice 1.4:

@ $x=1$

$$f(x) = \begin{cases} 3x^2+4x-5 ; & x>1 \\ x+11 ; & x<1 \end{cases}$$

$f(1)$ is not defined, so $f(x)$ is not cont. at $x=1$

@ $x=3$

$$g(x) = \begin{cases} 4x-2 ; & x \leq 3 \\ 3-x^2 ; & x>3 \end{cases}$$

$g(3)=10$, but $\lim_{x \to 3} g(x) = DNE$, so $g(x)$ is not continuous at $x=3$

@ $x=3$

$$i(x) = \begin{cases} x^2-2x+2 ; & x<3 \\ 5 ; & x=3 \\ 5x-10 ; & x>3 \end{cases}$$

$i(3)=5$, $\left(\lim_{x \to 3^+} i(x)=5\right) = \left(\lim_{x \to 3^-} i(x)=5\right)$, and

$i(3) = \lim_{x \to 3} i(x) = 5$, so $i(x)$ is continuous at $x=3$

Doesn't have to be piecewise to be discontinuous*

make it continuous

$$f(x) = \begin{cases} cx^2-3 & x \leq 2 \\ cx+2 & x>2 \end{cases}$$

$$\lim_{x \to 2^-} cx^2-3 = \lim cx+2$$

$$4c-3 = 2c+2$$

$$c = 5/2$$

$$f(x) = \begin{cases} ax+b & x>3 \\ 2bx-a & x<3 \\ 2 & x<3 \end{cases}$$

$\lim_{x \to 3^-} 2bx-a = 2$ $\lim_{x \to 3^+} ax+b = 2$

$6x-a$ $ax+6$

Discontinuity

1. Holes: Point disconuity
   $\frac{x^2-4}{x-2}$ Removable discontinuity

2. Step: Non-removable discontinuity

3. VA: asymptote discontinuity (non-removal)